



UNIVERSIDAD DE LAS FUERZAS ARMADAS “ESPE”

DEPARTAMENTO DE CIENCIAS DE LA COMPUTACIÓN

CARRERA INGENIERÍA DE SOFTWARE

Aplicaciones distribuidas - NRC 30732

Tema:

Evaluación Parcial

Profesor:

Ing. Paulo Galarza Cesar Sanchez

Nombre:

MOLINA GUAICO GUSTAVO STIVEN

Fecha:

22 de Junio del 2026

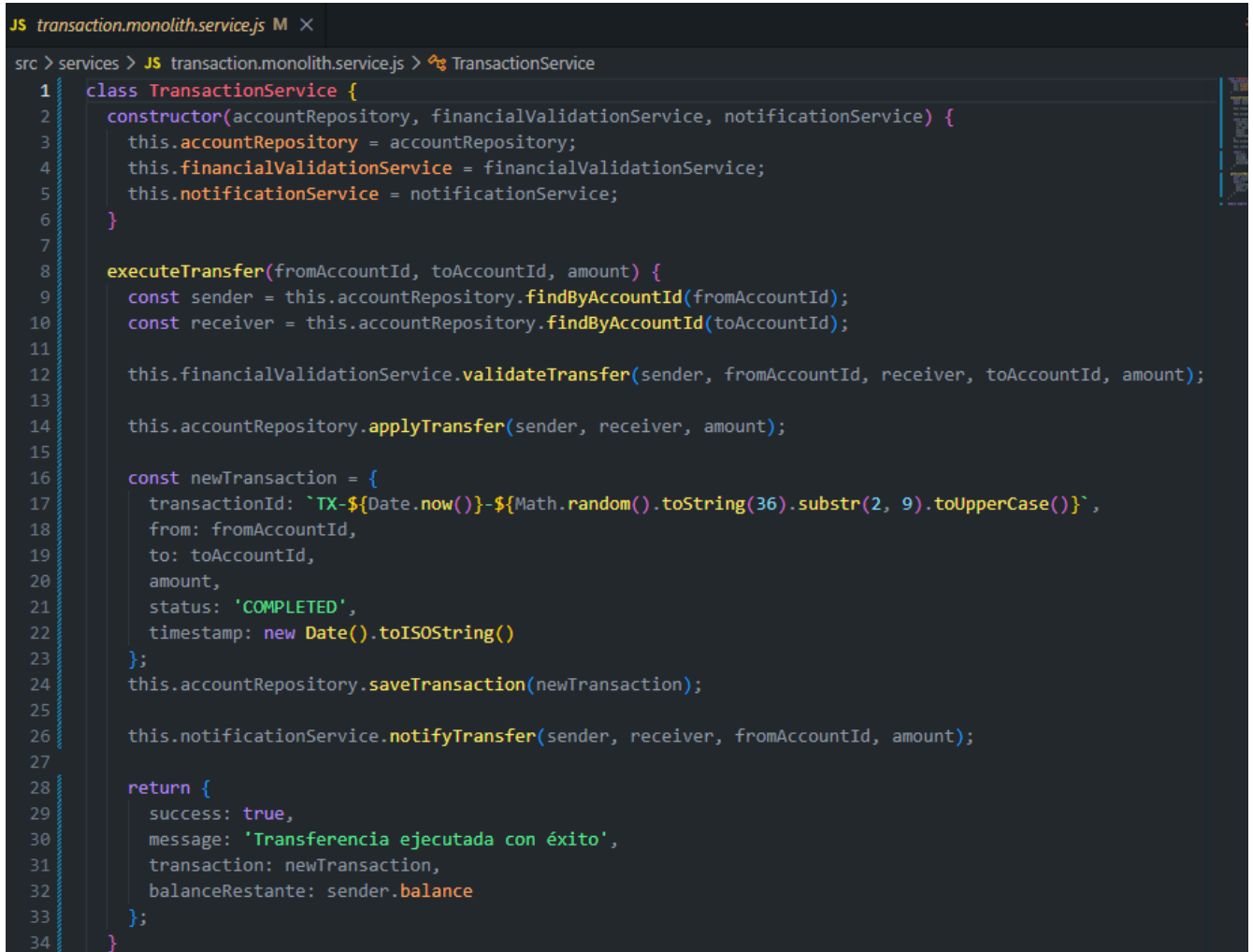
1. Desarrollo

1.1. Inyección de Control y Responsabilidad Única

1.1.1. Creación de la rama feature/01-refactor-solid.

```
PS C:\Users\steve\OneDrive\Escritorio\7mo\Distribuidas\Parcial 2\fintech-securepay-base> git checkout -b feature/01-refactor-solid
Switched to a new branch 'feature/01-refactor-solid'
PS C:\Users\steve\OneDrive\Escritorio\7mo\Distribuidas\Parcial 2\fintech-securepay-base>
```

1.1.2. Descomposición del monolito aplicando SRP



```
JS transaction.monolith.service.js M X
src > services > JS transaction.monolith.service.js > TransactionService
1 class TransactionService {
2   constructor(accountRepository, financialValidationService, notificationService) {
3     this.accountRepository = accountRepository;
4     this.financialValidationService = financialValidationService;
5     this.notificationService = notificationService;
6   }
7
8   executeTransfer(fromAccountId, toAccountId, amount) {
9     const sender = this.accountRepository.findByAccountId(fromAccountId);
10    const receiver = this.accountRepository.findByAccountId(toAccountId);
11
12    this.financialValidationService.validateTransfer(sender, fromAccountId, receiver, toAccountId, amount);
13
14    this.accountRepository.applyTransfer(sender, receiver, amount);
15
16    const newTransaction = {
17      transactionId: `TX-${Date.now()}-${Math.random().toString(36).substr(2, 9).toUpperCase()}`,
18      from: fromAccountId,
19      to: toAccountId,
20      amount,
21      status: 'COMPLETED',
22      timestamp: new Date().toISOString()
23    };
24    this.accountRepository.saveTransaction(newTransaction);
25
26    this.notificationService.notifyTransfer(sender, receiver, fromAccountId, amount);
27
28    return {
29      success: true,
30      message: 'Transferencia ejecutada con éxito',
31      transaction: newTransaction,
32      balanceRestante: sender.balance
33    };
34  }
```

1.1.3. Aislamiento de la lógica financiera, el almacenamiento de estados y las notificaciones por consola en servicios de bajo nivel independientes.

```

JS account.repository.js U X
src > services > JS account.repository.js > ...
1  const usersDb = [
2    { id: 'usr_001', email: 'estudiante.alpha@espe.edu.ec', accountAlpha: 'ACC-12345', balance: 1500.00 },
3    { id: 'usr_002', email: 'docente.beta@espe.edu.ec', accountAlpha: 'ACC-67890', balance: 350.50 }
4  ];
5
6  const transactionsHistory = [];
7
8  class AccountRepository {
9    findById(accountId) {
10     return usersDb.find(u => u.accountAlpha === accountId) || null;
11   }
12
13   applyTransfer(sender, receiver, amount) {
14     sender.balance -= amount;
15     receiver.balance += amount;
16   }
17
18   saveTransaction(transaction) {
19     transactionsHistory.push(transaction);
20   }
21
22   getTransactionsHistory() {
23     return [...transactionsHistory];
24   }
25 }
26
27 module.exports = new AccountRepository();

```

1.1.4. Implementación de inversión de dependencias (DIP)

- Nivel 1 (El servicio recibe sus dependencias por el constructor)

```

services > JS transaction.monolith.service.js > transactionService
class TransactionService {
  constructor(accountRepository, financialValidationService, notificationService) {
    this.accountRepository = accountRepository;
    this.financialValidationService = financialValidationService;
    this.notificationService = notificationService;
  }
}

```

- Nivel 2 (El controlador recibe el servicio por el constructor)

```

class TransferController {
  constructor(transactionService) {
    this.transactionService = transactionService;
  }
}

```

- Nivel 3 (El cableado real ocurre en rutas)

```

const transactionService = new TransactionService(accountRepository, financialValidationService, notificationService);
const transferController = new TransferController(transactionService);

```

1.1.5. Merge controlado hacia la rama main

refactor(solid): segregar logica del monolito e inyectar dependencias... #1

Merged gsMolina02 merged 1 commit into main from feature/01-refactor-solid now

Conversation 0 Commits 1 Checks 0 Files changed 8 +168 -139

gsMolina02 commented now

... por constructor

Mention @copilot in a comment to make changes to this pull request.

refactor(solid): segregar logica del monolito e inyectar dependencias... a67b3ae

gsMolina02 merged commit eas15fb into main now

Pull request successfully merged and closed

You're all set — the feature/01-refactor-solid branch can be safely deleted.

Delete branch

Add a comment

Write Preview

Add your comment here...

Markdown is supported Paste, drop, or click to add files

Comment

Remember, contributions to this repository should follow our GitHub Community Guidelines.

1 participant

1.2. Criptografía y Flujo Distribuidor

1.2.1. Creación de la rama feature/02-auth-jwt

```
PS C:\Users\steve\OneDrive\Escritorio\7mo\Distribuidas\Parcial 2\fintech-securepay-base> git checkout -b feature/02-auth-jwt
Switched to a new branch 'feature/02-auth-jwt'
PS C:\Users\steve\OneDrive\Escritorio\7mo\Distribuidas\Parcial 2\fintech-securepay-base>
```

1.2.2. Ejecución del script [./keypair.sh](#)


```

try {
  const decodedToken = jwtService.verifyToken(token);
  req.user = decodedToken;
  next();
} catch (error) {
  if (error.name === 'TokenExpiredError') {
    return res.status(401).json({
      error: 'Token expirado',
      message: 'El token JWT ha expirado. Por favor, obtenga uno nuevo.'
    });
  }
  return res.status(403).json({
    error: 'Token inválido',
    message: error.message
  });
}
}

```

1.2.5. Merge hacia la rama main

feat(jwt): implementar firmado asimetrico rs256 y middleware de valid... #2

Merged gsMolina02 merged 1 commit into main from feature/02-auth-jwt now

Conversation 0 Commits 1 Checks 0 Files changed 2

gsMolina02 commented now

...acion autonoma public-key

Mention @copilot in a comment to make changes to this pull request.

feat(jwt): implementar firmado asimetrico rs256 y middleware de valid... 2c7268c

gsMolina02 merged commit 3f5812c into main now

Pull request successfully merged and closed

You're all set — the `feature/02-auth-jwt` branch can be safely deleted.

Delete branch

Add a comment

Write Preview

Add your comment here...

Markdown is supported Paste, drop, or click to add files

Close pull request Comment

1 participant

1.3. Gestión de Excepciones Lógicas vs. Errores Operacionales

1.3.1. Creación de la rama feature/03-observabilidad.

```
PS C:\Users\steve\OneDrive\Escritorio\7mo\Distribuidas\Parcial 2\fintech-securepay-base> git checkout -b feature/03-observabilidad
Switched to a new branch 'feature/03-observabilidad'
PS C:\Users\steve\OneDrive\Escritorio\7mo\Distribuidas\Parcial 2\fintech-securepay-base> |
```

1.3.2. Integración del SDK de monitoreo y creación del módulo `.src/instrument.js`

```
JS instrument.js U X
src > JS instrument.js > ...
1  require('dotenv').config();
2  const Sentry = require('@sentry/node');
3
4  Sentry.init({
5    dsn: process.env.SENTRY_DSN,
6    tracesSampleRate: 1.0,
7  });
8
```

1.3.3. Configuración del entorno de sentry

1.3.3.1. Creación del proyecto en sentry

Create a new project in 3 steps

Search Ctrl+K Ask Seer Ctrl+J

Filter Platforms

Popular Browser Server Mobile Desktop Serverless Gaming All

 NEXT.JS
  REACT
  REACT NATIVE
  NODE.JS
  LARAVEL
  FASTAPI
  FLUTTER
  DJANGO
  PYTHON
  EXPRESS
  BROWSER JAVASCRIPT
  PHP
  RAILS

 iOS
  NEXT.JS
  FLASK
  VUE
  ASP.NET CORE
  NUXT
  .NET MAUI
  ANGULAR
  ANDROID
  SPRING BOOT
  SYMFONY
  CLOUDFLARE WORKERS
  ELECTRON

 UNITY
  REMIX

2 Set your alert frequency

☒ Alert me on high priority issues

☐ When there are more than occurrences of one minute

☐ I'll create my own alerts later

☒ Notify via email

3 Name your project and assign it a team

Project slug Team

1.3.3.2. Conexión a la DNS de sentry

```
.env  X  .env.example M
.env
1  PORT=3000
2  PRIVATE_KEY_PATH=./private.pem
3  PUBLIC_KEY_PATH=./public.pem
4  SENTRY_DSN=https://85302f041f07e67754d33658c5bc638a@o4176.ingest.us.sentry.io/4511609150504960
5
```

1.3.3.3. Configuración de los controladores

- Error Lógico (No Alertar a Sentry)

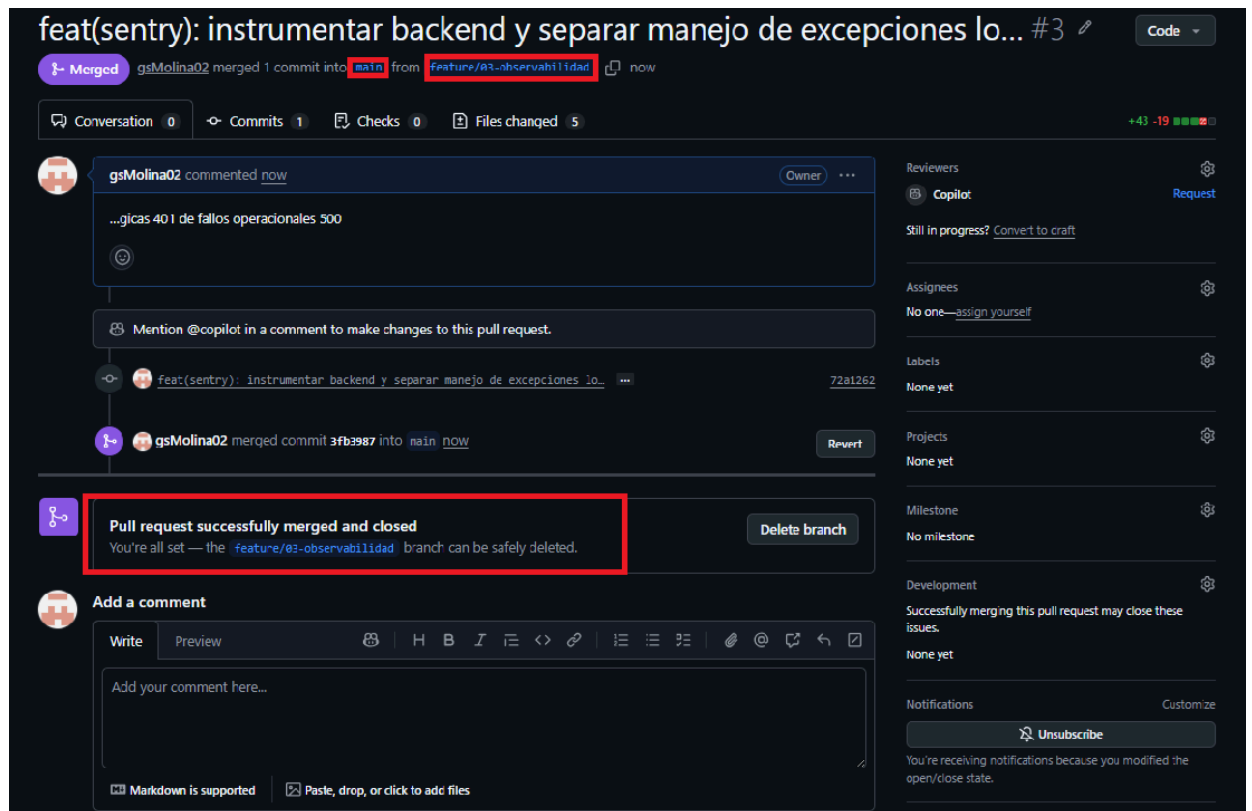
```
} catch (error) {
  if (error.name === 'TokenExpiredError') {
    return res.status(401).json({
      error: 'Token expirado',
      message: 'El token JWT ha expirado. Por favor, obtenga uno nuevo.'
    });
  }
  return res.status(403).json({
    error: 'Token inválido',
    message: error.message
  });
}
```

- Error Operacional (Alertar a Sentry)

```
} catch (error) {
  Sentry.withScope((scope) => {
    scope.setTag('user_id', req.user?.sub || 'anonymous');
    Sentry.captureException(error);
  });

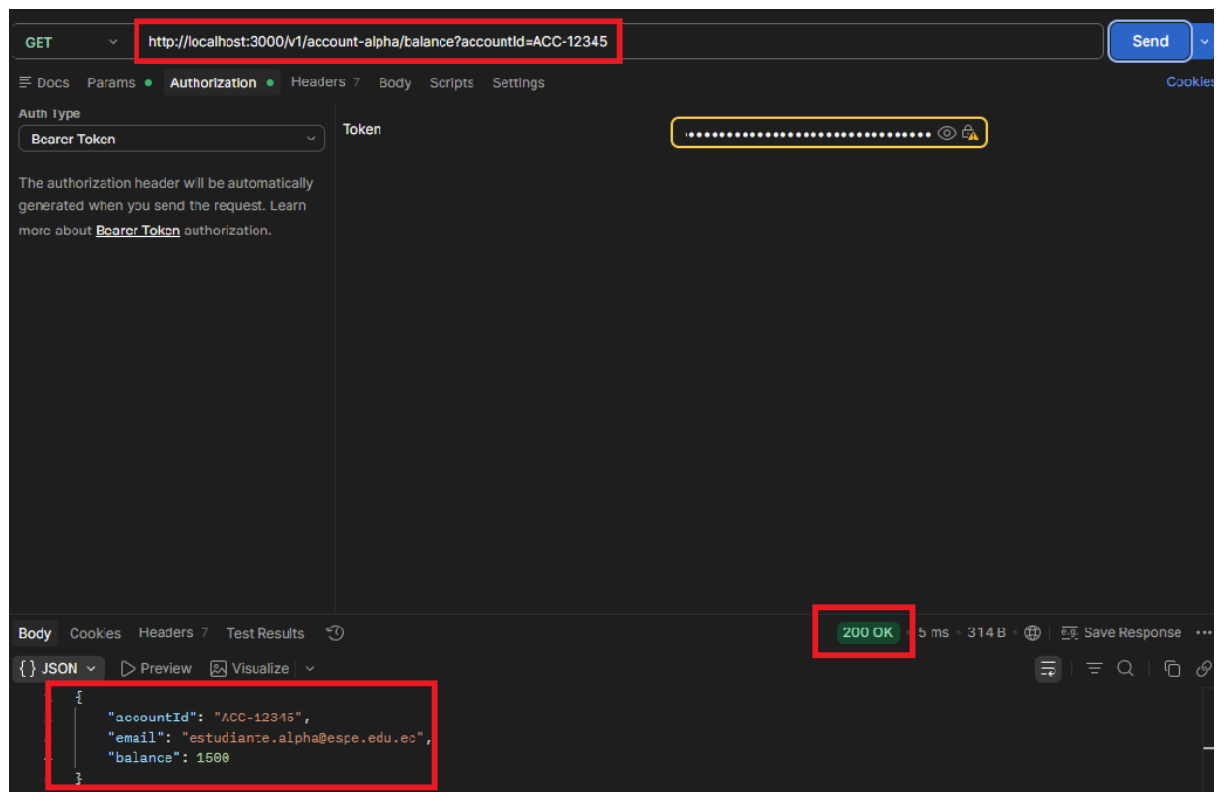
  return res.status(500).json({
    error: 'Error operacional',
    message: error.message
  });
}
```

1.3.4. Merge hacia la rama main

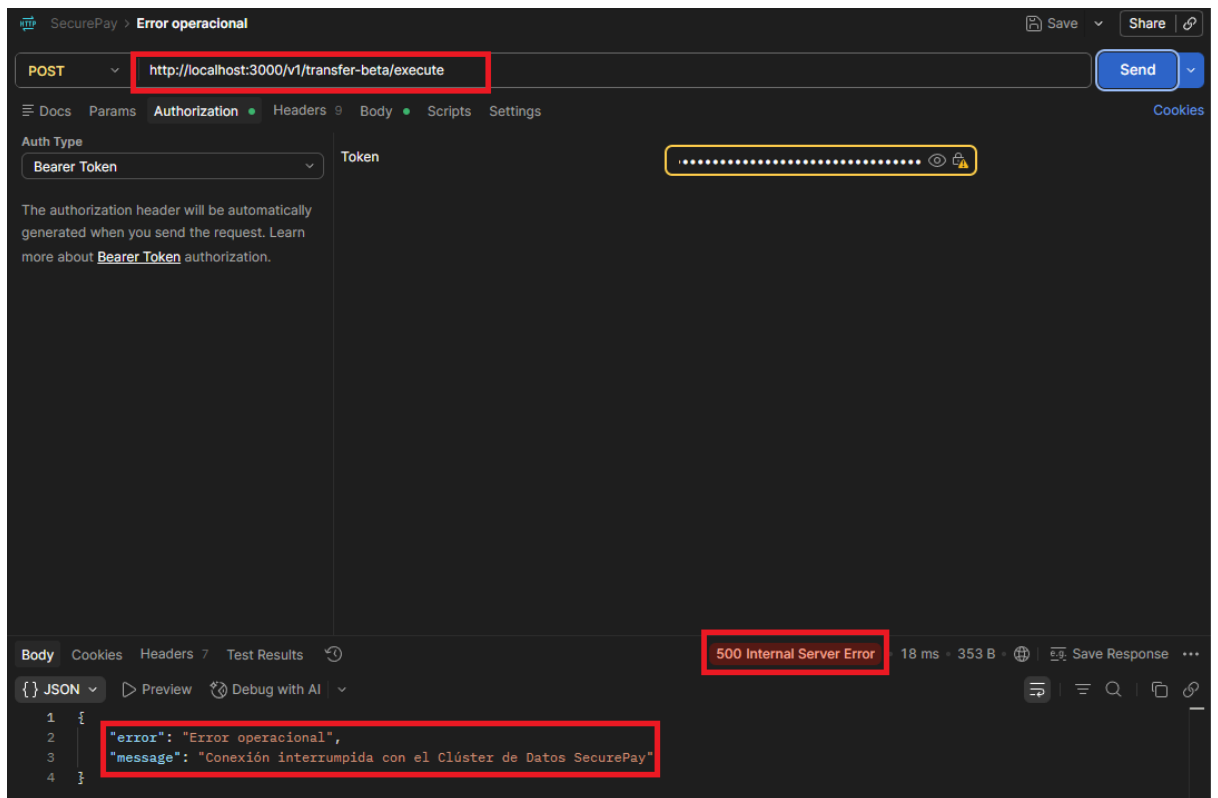


2. PostMan capturas

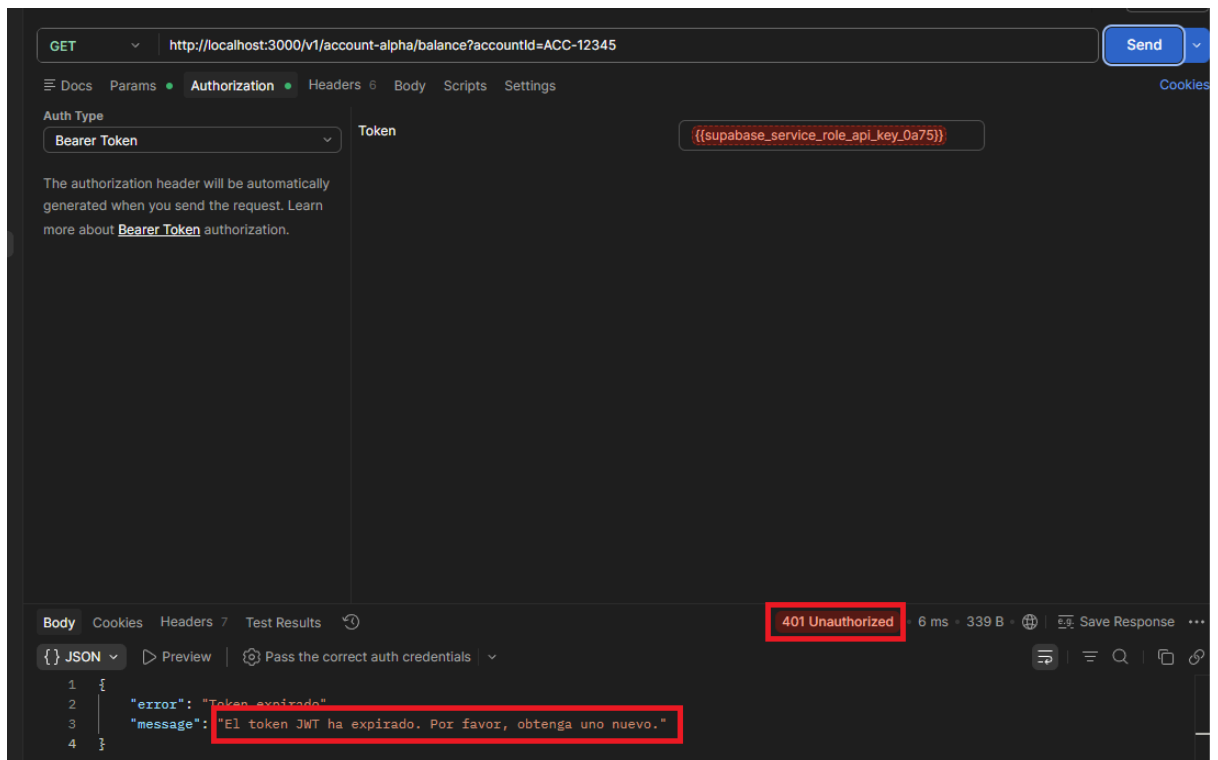
2.1. Acceso valido Alpha (200)



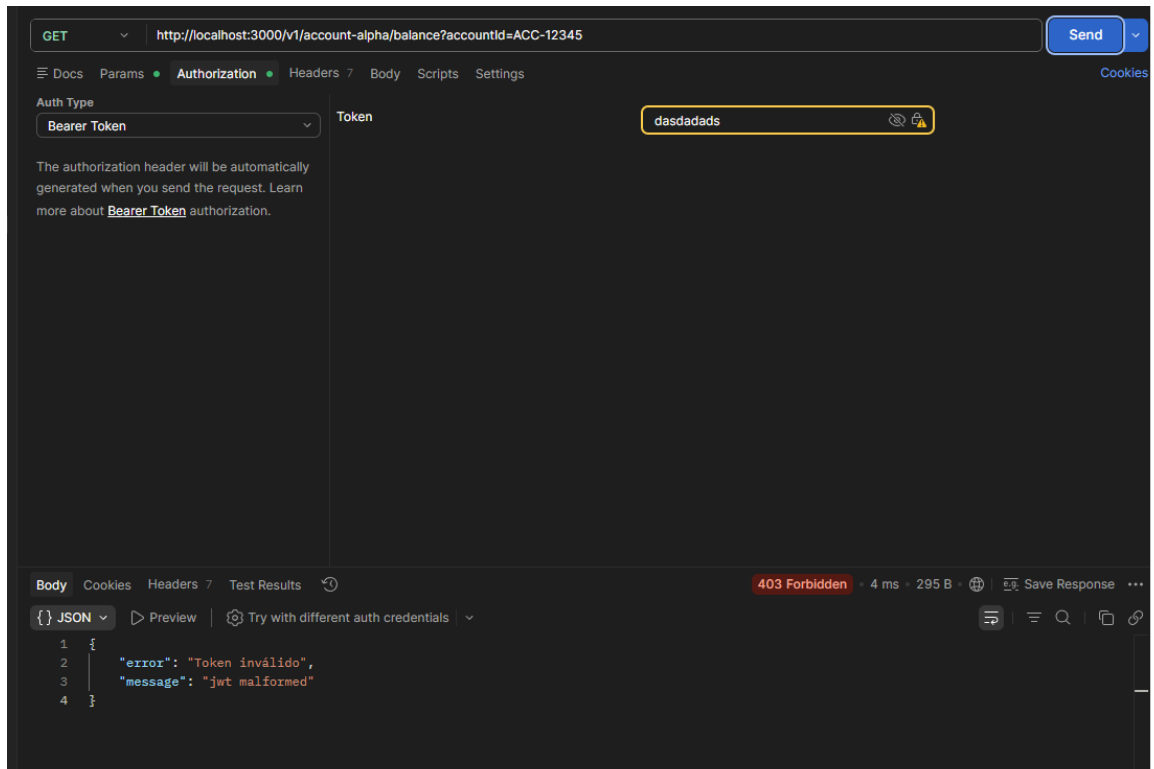
2.2. 02-acceso-valido-alpha



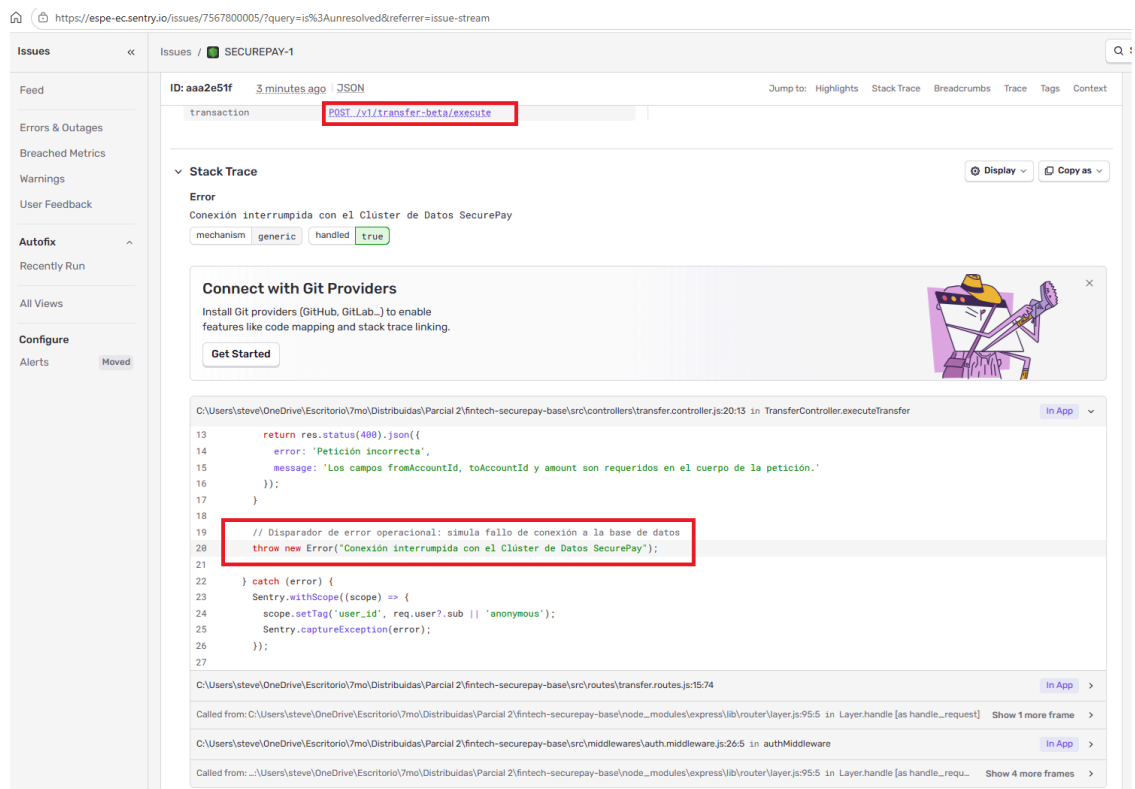
2.3. 03-token-expirado-401



2.4. 04-token-invalido-403



2.5. 05-sentry-error-500



2.6. 06-sentry-tag-user-id

Tags

browser	PostmanRuntime 7.54.0
name	PostmanRuntime
environment	production
handled	yes
interface_type	exception
level	error
mechanism	generic
os	Windows 10.0.26200
name	Windows

AllCustomApplicationClientOther

runtime	node v24.14.1
name	node
server_name	Steven
transaction	POST /v1/transfer-beta/execute
url	http://localhost:3000/v1/transfer-beta/execute
user_id	usr_001

Contexts

User

Geography Quito, Ecuador (EC)

Browser

Name PostmanRuntime

Version 7.54.0

Runtime

Name node

Version v24.14.1

Operating System

Name Windows

Version 10.0.26200

App

Free Memory 15.6 GiB

Memory Usage 88.0 MiB

Start Time 2026-06-22T12:59:19.359Z (12 minutes before this event)

Culture

Locale es-ES

Timezone America/Guayaquil

Device

Architecture x64

Boot Time 2026-06-15T15:36:55.908Z (7 days before this event)

CPU Description Intel(R) Core(TM) i7-14650HX

Free Memory 15.6 GiB

Memory Size 31.6 GiB

Processor Count 24

Processor Frequency (MHz) 2419

Trace Details

Client Sample Rate 1

Span ID 84dcad89c4e7ad09

Status unknown

Trace ID 5a731a340ede4ef733815315975b4d54

2.7. 07-postman-500-operacional

SecurePay > New Request

POST

http://localhost:3000/v1/transfer-beta/execute

Send

Docs Params Authorization Headers Body Scripts Settings

Auth Type

Bearer Token

Token

.....

The authorization header will be automatically generated when you send the request. Learn more about [Bearer Token](#) authorization.

Body Cookies Headers Test Results

500 Internal Server Error

7 ms · 353 B · Save Response

JSON

Preview Debug with AI

```

1 {
2   "error": "Error operacional",
3   "message": "Conexión interrumpida con el Clúster de Datos SecurePay"
4 }

```

3. Repositorio

[gsmolina02/SecurePay](https://github.com/gsmolina02/SecurePay)